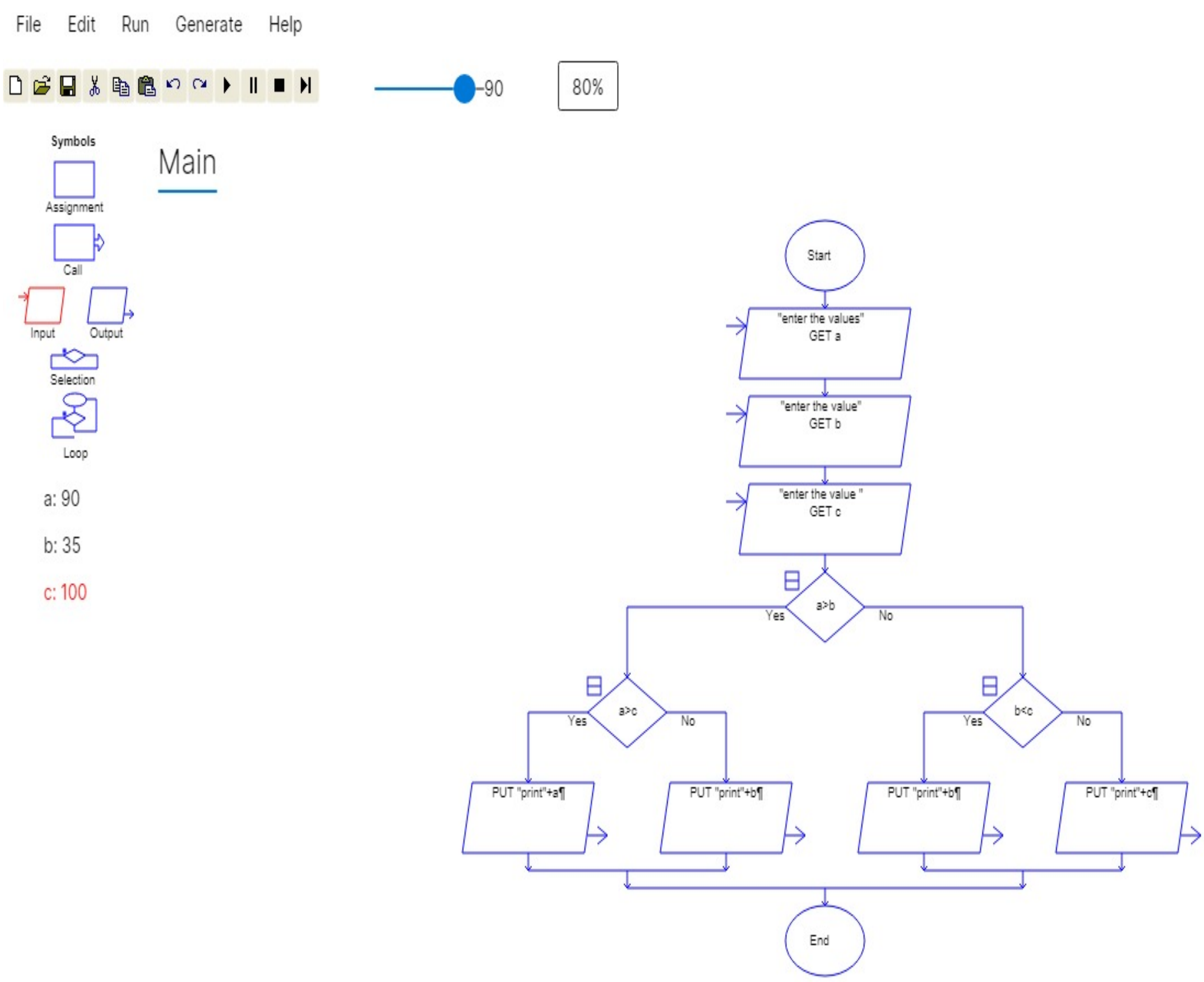
RAPTOR PROGRAMS

1. Draw a raptor flowchart to read three integer numbers (a, b, and c) from the user and determine the largest among them using decision symbols. Display the maximum number as the output.

AIM:

To develop a raptor flowchart that reads three numbers from the user and determines the maximum among them using decision-making statements.

RAPTOR FLOWCHART:

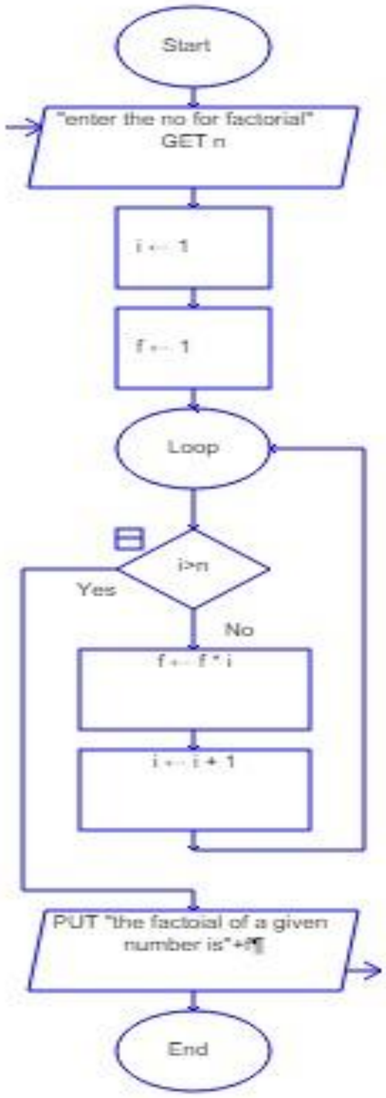


2. Using raptor, draw and validate a flowchart to find the factorial of a given positive integer. The program should read an integer N, calculate its factorial using a loop, and display the result.

AIM:

To draw and validate a RAPTOR flowchart for finding the factorial of a given positive integer using an iterative approach.

RAPTOR FLOW:

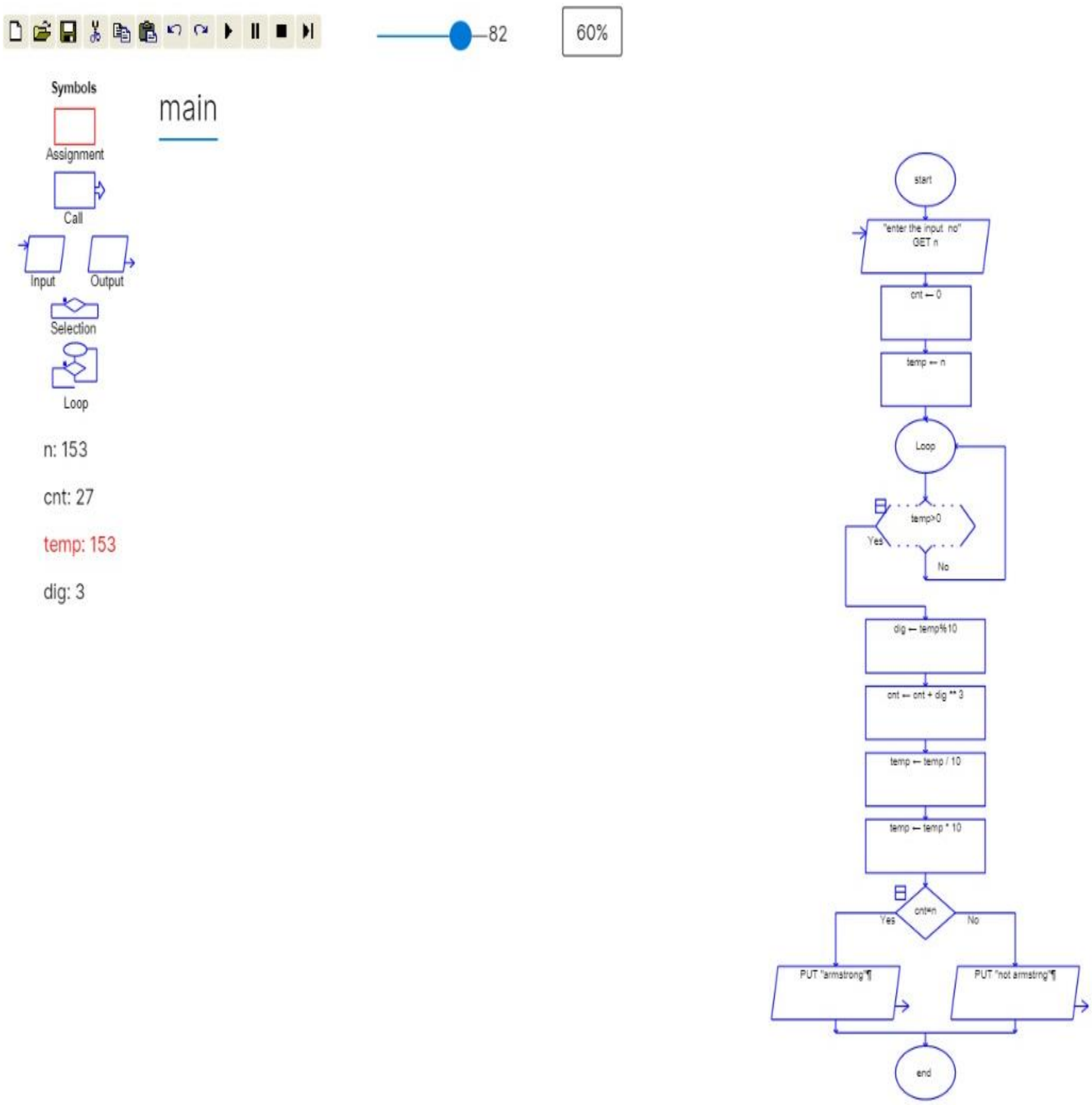


3. Using raptor, draw and validate a flowchart to check whether a given positive integer is an armstrong number or not. The program should read an integer N, calculate the sum of the cubes of its digits, compare the sum with the original number, and display whether the number is an armstrong number or not.

AIM:

To draw and validate a RAPTOR flowchart for checking whether a given positive integer is an Armstrong number using loops and conditional statements.

RAPTOR FLOWCHART:

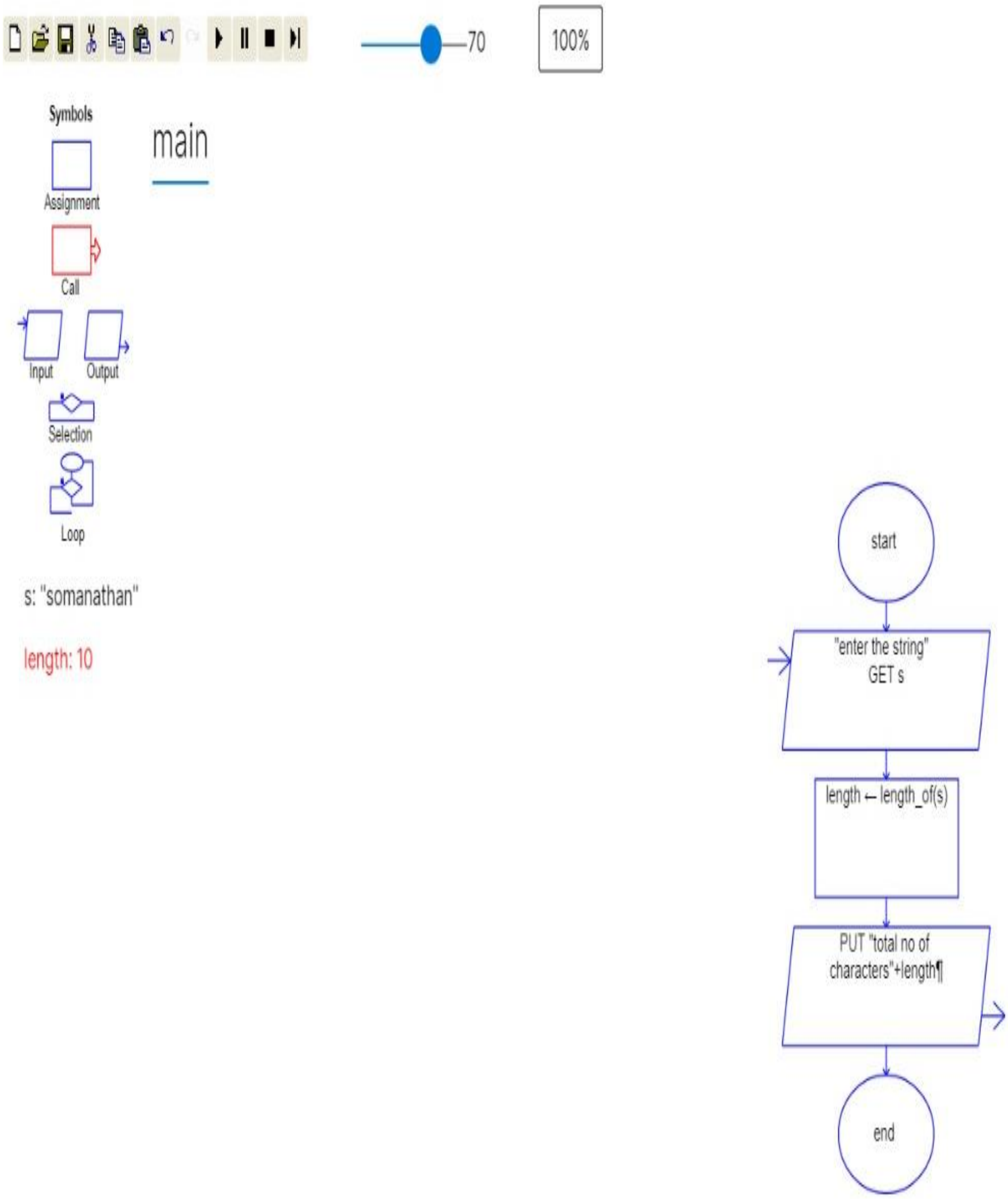


4. The string is a sequence of characters placed in double quotes (" "). Performing different operations on string data is called String Handling. Strings are immutable. Whenever a change to a String is made, an entirely new String is created. If we want to store a group of characters we can use a char array. String provides various methods to perform different operations on strings. Using Raptor draw a flowchart to display the total number of characters in the string and returns it.

AIM:

To draw and validate a RAPTOR flowchart to find and display the total number of characters present in a given string using the string length function.

RAPTOR FLOWCHART:

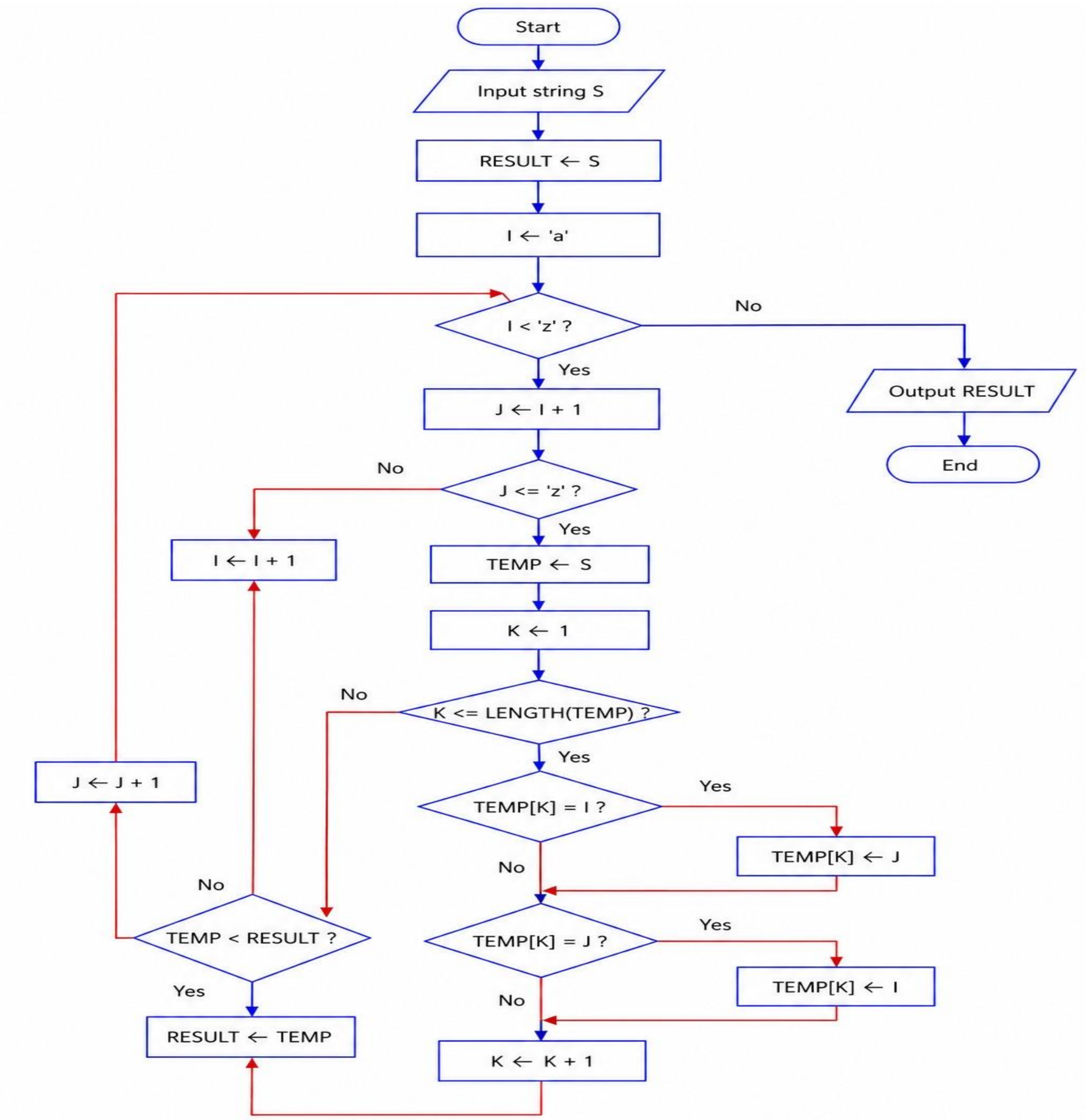


5. Using Raptor – Draw and validate a flowchart for a given string of lower case English alphabets. One can choose any two characters in the string and replace all the occurrences of the first character with the second character and replace all the occurrences of the second character with the first character. Using Raptor draw and validate the flowchart lexicographically smallest string that can be obtained by doing this operation at most once.

AIM:

To draw and validate a RAPTOR flowchart to obtain the lexicographically smallest string by swapping all occurrences of any two lowercase English alphabet characters in a given string at most once.

RAPTOR FLOWCHART;

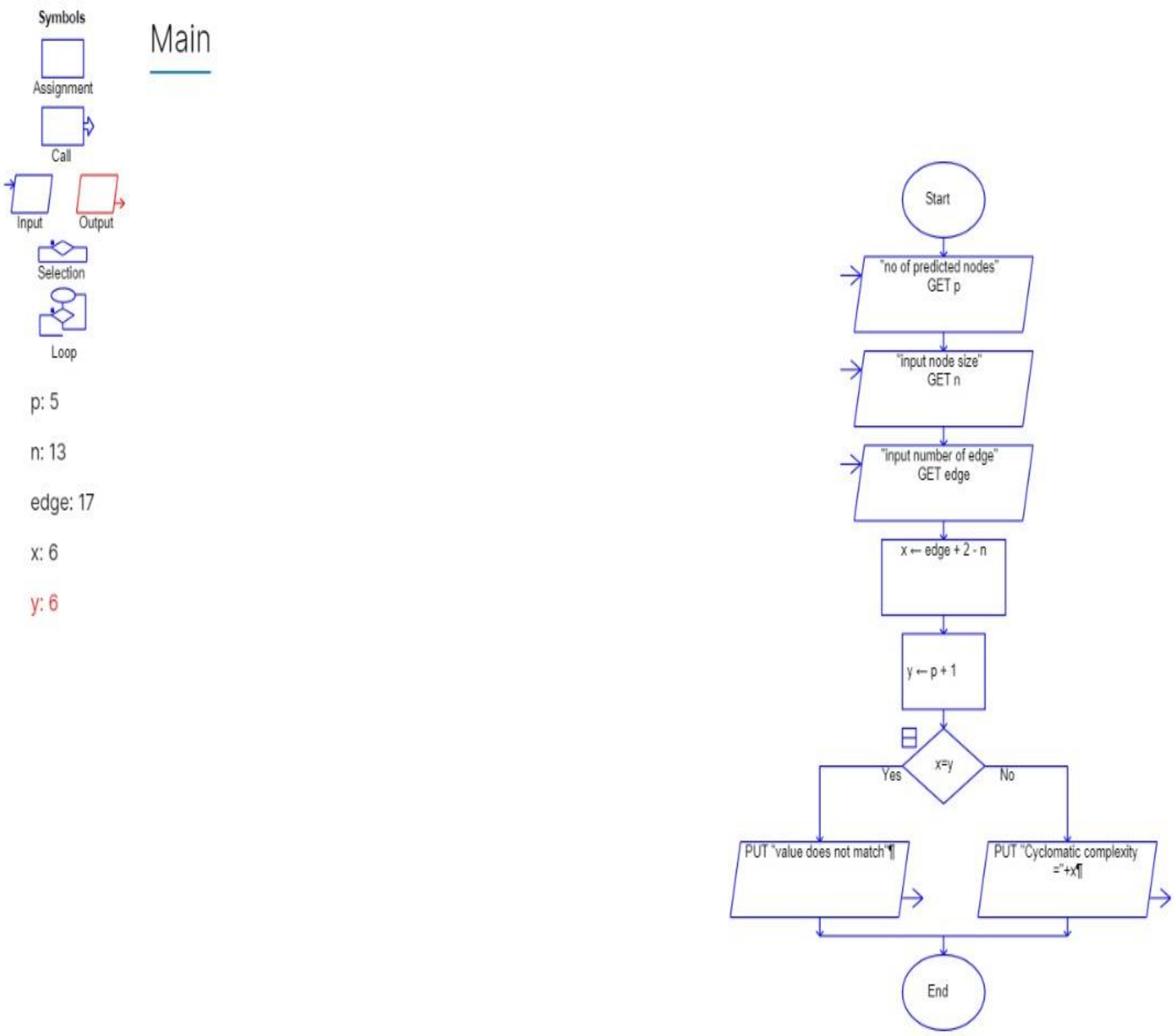


6. Cyclomatic Complexity in Software Testing is a testing metric used for measuring the complexity of a software program. It is a quantitative measure of independent paths in the source code of a software program. Cyclomatic complexity can be calculated by using control flow graphs or with respect to functions, modules, methods or classes within a software program. Cyclomatic complexity for a flow graph G is V(G)=E-N+2. Cyclomatic complexity is a software metric (measurement), used to indicate the complexity of a program, Cyclomatic complexity = E - N + P where, E = number of edges in the flow graph, N = number of nodes in the flow graph, P = number of nodes that have exit points. Find Cyclomatic Complexity for a graph having number of edges as 17, number of nodes as 13 and number of predicate nodes in the flow graph as 5.

AIM:

To draw and validate a RAPTOR flowchart to calculate the Cyclomatic Complexity of a flow graph using the given number of edges, nodes, and predicate nodes, and display the result.

RAPTOR FLOWCHART:



Result

Thus, the RAPTOR flowcharts for the given problems were successfully designed, validated, and executed. The algorithms produced the expected outputs for all test cases, demonstrating the correct implementation of programming concepts such as arithmetic operations, decision making, looping, string handling, and software metrics using RAPTOR.